One barrier or two steps? Nucleation statistics for gypsum and how additives change its pathway

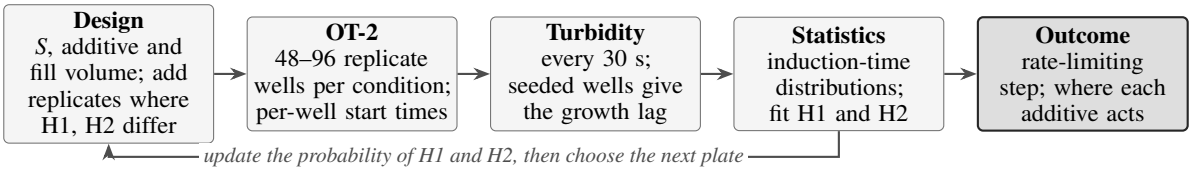
Sterling G. Baird (PI), Assistant Professor of Mechanical Engineering, Brigham Young University
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Research question. When gypsum ($CaSO_4{\cdot}2H_2O$) crystallizes from water, is the rate set by a single nucleation barrier, as classical nucleation theory assumes, or by forming and then transforming a precursor? And when an additive delays crystallization, which step does it act on?

Knowledge gap. Gypsum is a model ionic crystal grown from water, and a costly scale in geothermal, oil and gas, and water systems. Its induction times have been mapped against supersaturation, salinity and temperature, one onset per condition. Scattering and imaging show nanoscale precursors [1,2] but have not linked the precursor step to the rate. Nucleation is stochastic, so that answer lies in the distribution of induction times over many identical samples [3,4], yet scale and additive studies usually report one onset time per condition.

Hypotheses. *H1, single barrier:* each well crystallizes by one rare event, so the fraction of wells still clear decays exponentially after a fixed growth lag, $\ln J$ falls linearly with $1/\ln^2 S$, and an additive moves only the slope (interfacial energy) or intercept (prefactor) of that line. *H2, precursor-mediated:* two slow steps in series give a sigmoidal, non-exponential distribution, a break in slope where the slower step switches (unlike a wall-to-bulk switch, with no change in volume scaling), and additives that change the distribution's shape, not just its scale.

Approach. An Opentrons OT-2 fills 96-well plates from filtered $CaCl_2$ and Na_2SO_4 stocks to set the supersaturation S (from Pitzer activities), ionic strength and additive (a phosphonate, a polyacrylate), with 48–96 replicate wells per condition and a logged start time for each well. Turbidity is read every 30 s by a plate reader, or by the lab's camera under fixed lighting. Fill volume and wall material are varied too: rates that scale with volume mean nucleation in the bulk, and rates that follow the wall mean the walls. Seeded, sealed wells measure the growth and detection lag directly, so a spread in lag (which additives that slow growth can cause) is not mistaken for H2. Selected conditions suit time-resolved SAXS at a BES user facility. Replicates are added where H1 and H2 predict the most different distributions; the automation is the means, not the aim.



Expected outcomes. (1) Nucleation rates with uncertainties across S, ionic strength and additives, with bulk and wall contributions separated; (2) a test of whether precursor formation controls the rate; (3) which step each additive acts on. Together these give a statistically grounded time–supersaturation–transformation map for a model ionic crystal, and a mechanistic basis for choosing additives.

Fit and resources. The project asks why crystallization proceeds as it does, through the roles of *nucleation* and *precursor and intermediate states* in materials synthesis. It does not optimize a process or develop automation for its own sake. The PI works on self-driving labs and Bayesian experimental design; the lab's OT-2 already runs turbidity-onset precipitation series, so preliminary distributions can be measured before an application.

References. [1] A. E. S. Van Driessche *et al.*, *Science* 336, 69 (2012), doi:10.1126/science.1215648. [2] T. M. Stawski *et al.*, *Nat. Commun.* 7, 11177 (2016), doi:10.1038/ncomms11177. [3] S. Jiang, J. H. ter Horst, *Cryst. Growth Des.* 11, 256 (2011), doi:10.1021/cg101213q. [4] T. Brandel, J. H. ter Horst, *Faraday Discuss.* 179, 199 (2015), doi:10.1039/C4FD00230J.